Using a Chatbot to Provide Formative Feedback: A Longitudinal Study of Intrinsic Motivation, Cognitive Load, and Learning Performance

Jiaqi Yin, Tiong-Thye Goh, and Yi Hu

***Abstract*—This study aimed to examine sustainable effects of chatbot-based formative feedback on intrinsic motivation, cognitive load, and learning performance. A longitudinal quasi-experimental design with 173 undergraduate students was conducted. The experiment is a between-subject design. Students either received formative feedback from a chatbot or a teacher. Utilizing linear mixed model and t-test for data analysis, results showed the following. First, chatbot-based feedback resulted in increased learning interest, perceived choice, and value while decreasing perceived pressure over time. Second, chatbot-based feedback was effective in reducing cognitive load, particularly when learning contents involved conceptual or difficult knowledge. Finally, chatbot-based feedback was found to be more efficient and effective in supporting the mastery of application-based knowledge compared with teacher-based feedback. This study has practical implications for the design of chatbots, and it also enriches the methods of providing ongoing formative feedback in large-scale classrooms.**

***Index Terms*—Chatbot, cognitive load, formative feedback, intrinsic motivation, learning performance.**

I. Introduction

FORMATIVE feedback aims to provide students with information that modifies their thinking or behavior to improve learning [1]. Specifically, formative feedback has the potential to close the gap between students' existing understanding or performance and their expected goals, thereby reducing cognitive load and enhancing learning performance [2]. Research showed that formative feedback provided by teachers in a classroom setting improved college students' performance and their abilities to solve complex problems [3]. However, it is often difficult for teachers to offer individualized feedback to every student, and there can be delays in providing feedback, particularly within a sizable classroom setting. Furthermore, effective feedback requires not only the teacher's participation but also the students' involvement [4]. To address these challenges, previous researchers proposed incorporating artificial intelligence (AI) technology to provide personalized and instant feedback, as well as create interactive learning materials (e.g., interactive quizzes or games) to increase student engagement [5].

AI chatbots, also called conversational agents, virtual assistants, or dialogue systems, find applications in various learning scenarios where they serve multiple purposes [6]. They provide answers to commonly asked questions and facilitate short quizzes [7]. While some research has incorporated feedback into their chatbots, their primary objective was not to focus on feedback design. Instead, they aimed to assess the chatbot's effectiveness in enhancing learning compared with nonchatbot situations. For example, Balderas et al. [8] proposed a chatbot-driven learning platform designed to support students in mastering structured query language (SQL). The interaction between students and the chatbot encompassed several steps, such as engaging in exercises, receiving hints, presenting solutions, seeking assistance, and receiving corrective feedback. The study's outcomes provided evidence that students who utilized the chatbot performed better on the final SQL exam than those who did not have access to it. Despite the demonstrated positive effects of chatbots as formative feedback tools on engagement and task completion [9], there is still a lack of research on their sustainable influence on learning performance and psychological factors. For example, Vijayakumar et al. [10] used SQL Quizbot, a chatbot deployed on Facebook Messenger for formative feedback. Through a cross-sectional survey, they found that immediate feedback was more advantageous to users than cumulative feedback. However, the effects of chatbot-based formative feedback on intrinsic motivation and cognitive load over an extended time have not been thoroughly investigated.

Research has shown that chatbots with their interactivity and multimodal features can enhance students' learning interest and alleviate learning pressure, thus promoting intrinsic motivation [11]. According to the self-determination theory (SDT) [12], students with higher intrinsic motivation pay more attention to learning and produce better learning outcomes. In addition,

Manuscript received 3 May 2023; revised 9 September 2023 and 11 December 2023; accepted 1 February 2024. Date of publication 8 February 2024; date of current version 17 April 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 71942001 and in part by STI 2030—Major Projects under Grant 2021ZD0200500. *(Corresponding author: Yi Hu.)*

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by teachers from Hubei University and met ethical requirements.

Jiaqi Yin is with the Lab of Artificial Intelligence for Education, Shanghai Institute of Artificial Intelligence for Education, School of Computer Science and Technology, East China Normal University, Shanghai 200062, China (e-mail: 52215901018@stu.ecnu.edu.cn).

Tiong-Thye Goh is with the School of Information Management, Victoria University of Wellington, Wellington 6012, New Zealand (e-mail: Tiong.Goh@vuw.ac.nz).

Yi Hu is with the Shanghai Key Laboratory of Mental Health and Psychological Crisis Intervention, School of Psychology and Cognitive Science, East China Normal University, Shanghai 200062, China (e-mail: yhu@psy.ecnu.edu.cn).

Digital Object Identifier 10.1109/TLT.2024.3364015

the cognitive load theory (CLT) [13] suggests that ineffective instructional design needs to be avoided as it causes extraneous cognitive load and interferes with learning. Excessive cognitive load occurs when the presentation of information or the tasks required of learners are perplexing [14]. For instance, any instructional procedure that requires learners to engage in either a search for a problem solution or a search for referents in an explanation is likely to impose a heavy extraneous cognitive load (i.e., the explanation mentions Part B in Part A without clarifying where to locate Part B.) [13]. Educational chatbots that adopt good instructional practices that respond with appropriate feedback may overcome the occurrence of extraneous cognitive load.

Limited research has compared teacher-based feedback with technology-based feedback in the literature. Therefore, this study will adopt good design principles to develop an educational chatbot to provide formative feedback. To evaluate the effects over an extended period resulting from chatbot-based formative feedback interactions, we conducted a comparison study with repeated measures to address the following questions.

1) Do students who receive chatbot-based formative feedback show increased interest, perceived choice, competence, and value over time?
2) Do students who receive chatbot-based formative feedback show decreased stress over time?
3) Do students who receive formative feedback from chatbots have a lower cognitive load?
4) Do students who receive formative feedback from chatbots have better learning performance?

II. Related Works

A. Formative Feedback

Formative feedback aims to increase student knowledge, skills, and understanding and not to provide grades or overall scores [2]. Past literature has focused primarily on teacher-based and technology-based feedbacks. Teacher-based formative feedback included theoretical and empirical research. Theoretical research is involved in what teachers need to implement feedback well. For example, Boud and Dawson [15] developed a teacher feedback literacy competency framework. They identified three levels of teacher feedback literacy. The macro level focused on the program design and development. The meso level focused on the course module design and implementation, and the micro level focused on the feedback practices relating to individual student assignments. Empirical research is dedicated to exploring the impact of different types of feedback on students' learning performance and psychological factors. For example, Patra et al. [16] studied how teachers' written and verbal feedback affects English as a foreign language learners' academic anxiety, performance, and learning attitude. They found that feedback improved academic performance and reduced anxiety. In addition, Olsen and Hunnes [17] assessed how the students respond to being taught academic writing in two different feedback modalities provided by the teacher simultaneously: face-to-face interaction and electronic annotations. They found that students were equally content with both methods. However, the sustainability of combining the two modalities is uncertain due to heavy workloads and time constraints.

In terms of technology-based formative feedback, relevant research focuses on the impact of different technologies or feedback strategies on learning performance. For instance, Knight et al. [18] developed an open-source AcaWriter tool. This tool offers feedback on rhetorical techniques and has been shown to enhance student writing. Furthermore, Hao et al. [19] designed different automated formative feedback systems for programming assignments. These systems encompass knowledge of results, knowledge of correct responses, and elaborated feedback (EF). The results found that students in EF group have better performance. In a study, an adaptive immediate feedback system driven by AI was introduced. This system utilizes a hybrid data-driven algorithm to give students real-time updates on programming progress and code accuracy. This approach led to enhanced student performance [20].

In general, teacher-based feedback lacks personalization, immediacy, and interactive elements [21], [22]. In addition, high-quality feedback in a large class context can be difficult to disseminate in a timely and consistent fashion [23]. Technology-based feedback facilitates the implementation of formative feedback in large classes. Furthermore, with the increasing sophistication of chatbots, there is a potential to address the above dilemma as chatbots act as personal tutors who can interact with each student in the large class context.

B. Chatbot-Based Learning

Chatbots emulate human dialogues through voice or text interactions. Existing studies predominantly view chatbots as educational tools for addressing students' queries. The primary research emphasis lies in enhancing learning outcomes, including language acquisition [24], academic advancement [25], and collaborative learning [26]. For example, Essel et al. [27] used a chatbot that automatically responds to a student's question. They found that the students who interacted with the chatbot performed better academically than those who interacted with the course instructor.

Recent studies on educational chatbots have also focused on their impact on students' subjective experiences [28]. For instance, Zhang et al. [29] presented a chatbot framework incorporating context recognition, learning experiences, and emotion management to bolster learners' emotional confidence. Similarly, Fidan and Gencel [30] employed chatbots and peer feedback mechanisms within instructional videos to enhance intrinsic motivation among preservice teachers in online learning. Likewise, a study aimed to create and assess a chatbot's impact on enhancing nursing skills through nonface-to-face video lectures for nursing college students. The chatbot encompassed the introduction, main course, and conclusion phases. The introduction incorporated quizzes, multiple-choice, and open-ended questions to gauge prelearning comprehension. Students receive feedback at this stage, such as praise or correct answers, after each question. Findings revealed that the chatbot group, compared with the nonchatbot group, elevated students' interest and self-directed learning [31].

Previous studies have reported that chatbots improved learning performance and subjective experiences, such as intrinsic motivation [11]. However, prior findings might have been influenced by the short-term novelty effects, and there is a need to verify whether the chatbot approach has a sustainable impact on students' learning performance and subjective experiences, especially in formative feedback learning situations. Hence, we integrated a chatbot to provide ongoing formative feedback, tracking the changes in learning performance, intrinsic motivation, and cognitive load.

III. Design of Chatbot-Based Formative Feedback

We designed chatbot-based formative feedback following six design principles. The first design principle (DP1) and second design principle (DP2) are derived from SDT [32], and the third and fourth principles are drawn from CLT [13]. The remaining two are from the feedback framework of Narciss and Huth [2] and Hattie and Timperley [33].

First, SDT is an empirically derived theory of human motivation and personality in social contexts. It further differentiates between intrinsic and extrinsic motivation. An intrinsically motivated student participates in activities for their enjoyment or challenge, not for external rewards. Conversely, an extrinsically motivated student is driven by external factors, such as rewards or pressures, as described by Deci and Ryan [12]. SDT holds that intrinsic motivation is a crucial driver for learning [12]. Enhancing the students' perception of autonomy, competence, and relatedness can help increase their intrinsic motivation to learn. Autonomy refers to the need for self-initiated and selective actions. Competence refers to the perception that the individual has effectively performed a task confidently. Relatedness is defined as the affective support that an individual receives or gives to others during interactions. Second, CLT is an instructional theory that posits that the success of learning depends on the availability of adequate cognitive resources to meet the demands of learning activities. Cognitive load types include intrinsic, extraneous, and germane load. Intrinsic cognitive load is determined by the complexity of the task, while extraneous cognitive load is determined by the presentation form of content. CLT recommends minimizing extraneous cognitive load as much as possible, allowing more working memory to be utilized for learning [34]. Finally, the feedback framework of Hattie and Timperley [33] describes the primary components of feedback, including its content, function, and presentation. It further expounds that effective feedback should contain objectives, progress evaluation, and subsequent steps. Here are the specific design principles.

The DP1 is to utilize multimedia attributes and personalized characteristics of chatbots to enhance intrinsic motivation. We implement DP1 in our chatbot as follows: 1) provide feedback in the form of text, images, and emojis to engage students' learning interests and 2) offer varied feedback depending on students' responses, enabling them to decide whether to attempt the question again or receive the answer directly, thereby promoting autonomy.

The DP2 is to design some challenging questions to inspire students' perceived competence [35]. Intrinsic motivation can also be supported by challenges and interactive choices [36]. For instance, the learning chapters that require revision are presented in the form of questions. As per Bloom's teaching objectives, questions are designed from low to high cognitive levels.

The third design principle (DP3) involves deconstructing complex questions into manageable step-by-step tasks, thereby lessening cognitive stress. To put this principle into practice, students receive hints, analogies, explanations, examples, and similar aids upon answering incorrectly.

The fourth design principle (DP4) involves streamlining presentation. In our chatbot, we apply DP4 by eliminating nonessential elements and focusing solely on essential content. Presenting all information simultaneously could result in shallow processing and heightened cognitive load [37].

The fifth design principle (DP5) is to integrate evaluative and informative components into feedback [2]. The evaluative component confirms the correctness of the student's response, indicating whether it is accurate. The informative component including guiding information, such as hints, analogies, explanations, and examples, is provided when students answer incorrectly.

The sixth design principle (DP6) is not to immediately associate the content of detailed feedback with the correct answer or method that was given. Prior research shows that this association may result in low efficiency and superficial learning [38]. We implement this principle in our chatbot as follows. First, the chatbot provides students with positive evaluations, such as "Congratulations, that is right" (evaluative component) and short explanations about the answer (informative component) when they get the answer right. Likewise, the chatbot offers encouraging evaluations, such as "Wrong answer, do not be discouraged" and hints to support students' motivation to try again. In Chapter 1, a short explanation is not provided because the exercises require a series of calculation steps. If a student answers correctly, it means that each calculation step is correct, and there is no need for unnecessary information. However, Chapters 2 and 4 exercises involve connections between multiple knowledge points. Even if students answer correctly, there may be elements of guessing. Therefore, providing a brief explanation helps students verify if their problem-solving method is correct. Second, the chatbot offers the right answer without a detailed analysis unless students ask for it.

As for implementing chatbot-based formative feedback, first, we select the Flow.ai tool as our development platform. This tool encompasses the following key features:

1) inclusion of integrated machine learning algorithms;
2) visual creation of conversation flows;
3) a diverse array of dialogue components facilitating the generation of various question types;
4) instant one-click publishing capabilities.

The guides of Flow.ai tool can be obtained on Khoros' website [39].

Second, we implement a feedback mechanism by creating a knowledge base for the chatbot. The process involves presenting teacher-created exercises one at a time, followed by the chatbot

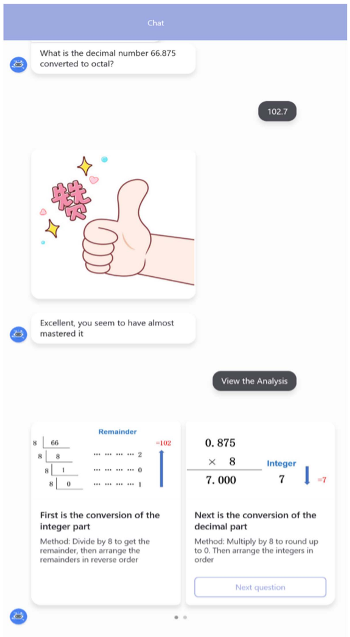

Fig. 1. Screenshot of the chatbot-based formative feedback conversation from Chapter 1.

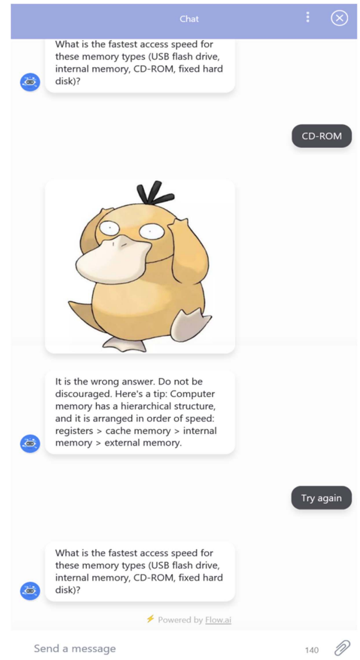

Fig. 2. Screenshot of the chatbot-based formative feedback conversation from Chapter 2.

receiving and comparing students' answers with the correct entries in the knowledge base. In addition, the chatbot can provide hints about the question, such as related knowledge points or the method and strategy of solving the question. After the chatbot provides a quick response, students can choose to view the answer or try again according to the prompt. After completing the last question, students can view their total scores by clicking "complete training." The screenshots of the chatbot session are shown in Figs. 1–3.

IV. Method

A. Participants

The participants were first-year undergraduates from a comprehensive university. A comprehensive university refers to an institution of higher learning with complete disciplines and a large-scale education. All participants enrolled in the basic university computer course. This study utilized a longitudinal quasi-experimental design with pretest and posttest at distinct time points. The participants were grouped randomly by the original class, which they were enrolled in. The teacher group comprised 109 law major students, whereas the chatbot group included 106 students majoring in international economics and trade (58%) and economics (42%). Among the 215 participants, 173 completed all surveys. The 42 respondents who did not complete all surveys were not included in the longitudinal analysis. Therefore, the final teacher group comprised 85 participants, while the chatbot group comprised 88 participants. Both groups were taught by the same teacher at different times on the same day, the chatbot group in the morning and the teacher group in the afternoon. During the review period, the students in the teacher group received feedback from the teacher, i.e., after the students had answered the questions independently, the teacher provided a unified explanation to the entire class. In the chatbot group, students practiced and received feedback from a chatbot. Specifically, the chatbot's capabilities encompassed delivering immediate feedback, enabling students to make multiple attempts at questions and review content at their own pace, as well as providing hints for tackling complex problems. Consider the following scenario. The teacher first introduced the chatbot, explaining its main functions and how it can aid the students in their review process. Then, the teacher shared the URL for the chatbot. Once the students accessed the chatbot, they started individually interacting with it. Initially, the chatbot introduced itself and asked for the students' names to record their exercise scores. Then, the chatbot presents questions, which can be multiple choice or fill in the blank. After the students answer, the chatbot verifies the answer and gives different feedbacks. For example, when a student fails to answer this question: what is the fastest access speed for these memory types? The chatbot provides a cartoon emoji, encouraging statements, and hints. Students can choose to attempt the question again based on these hints, as shown in Fig. 2.

Teacher-based feedback differs from chatbot-based feedback in the following ways.

1) Most of the feedback media are oral and pictures.
2) Feedback is given to the whole class, and students are not allowed to solve the question again.
3) For complex questions, the teacher will explain them all at once and will not provide hints.
4) The feedback session may be mixed with other interactions. For example, the teacher may ask other students to answer questions.

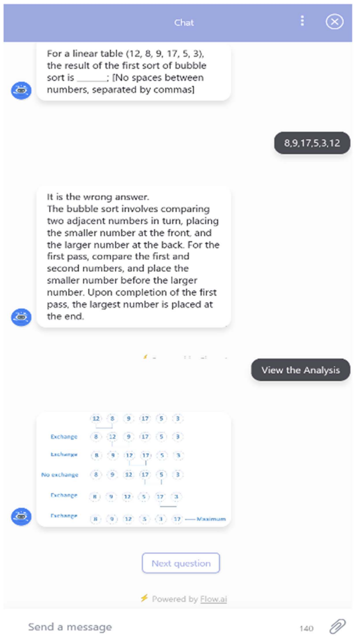

Fig. 3. Screenshot of the chatbot-based formative feedback conversation from Chapter 4.

5) The feedback session centers on the problem-solving process and does not involve student evaluation.

An example showcasing the teacher's unified explanation is provided below.

Question: How can the octal number 57.4 be converted to a binary number?

Teacher: Many students struggle with this question, so let me guide you through the calculation process. Converting octal numbers to binary involves transforming each octal digit into three binary digits. If there are fewer than three binary digits, we add leading zeros. For instance, let us consider the octal number 57.4. Converting 5 to binary results in 101, converting 7 to binary gives 111, and converting 4 to binary yields 100. It is important to note that this conversion follows the positional weight of 2. Thus, the final binary equivalent is 101111.100. (You can find the step-by-step calculations in a PowerPoint presentation for better clarity.)

B. Learning Materials

We chose the basic university computer course as our learning materials. We further selected the first chapter, "computer basic knowledge," the second chapter, "hardware fundamentals," and the fourth chapter, "algorithm and programming fundamentals" as our experiment content. Chapter 3 discusses the definition, history, and basic functions of operating systems since it is called "operating system basics." The learning contents are relatively easy for students to master. Therefore, it is not the subject of the review. This course focuses on Chapters 1, 2, and 4 and serves as the basis for the later in-depth study of computers, so this experiment selected these three chapters as experimental content. The course is a required course in many colleges and universities. The review questions for each chapter were designed based on the learning objectives, and a professor from the School of Computer Science was invited to develop exercises that covered all the learning objectives. A detailed description of the experiment materials and assessment is given in Table I.

C. Experimental Procedure

According to the definition of a longitudinal study, the feature is that two or more observations of the response variable are taken at different times. The occasions of measurement are not necessarily distributed evenly throughout the study [40]. Longitudinal studies can address fundamental questions concerning the assessment of within-individual changes in the response variable. Therefore, this study was conducted within 36 days. The learning units of Chapters 1, 2, and 4 were selected for evaluation. Chapter 1 is measured on day 1 and day 8, Chapter 2 is measured on day 15 and day 22, and Chapter 4 is measured on day 29 and day 36.

The first session of each chapter consists of two stages: a pretest and learning. The pretest phase for Chapter 1 includes both the intrinsic motivation scale and the prelearning knowledge test, while other chapters only have the prelearning knowledge test. During the learning phase, both groups of students are taught the chapter by the same teacher.

In the second session of each chapter, two stages occurred: review and posttest. These stages were scheduled seven days after the first session once all chapter content had been covered. During the review stage, the chatbot group answered questions posed by a chatbot, receiving scripted formative feedback. The teacher group utilized the "Questionnaire Star" platform for practice, followed by unified feedback from the teacher to the entire class.

The chatbot-based formative feedback process is explained in Section III. In contrast to the chatbot, the teacher initially selects challenging and error-prone questions based on experience. For the first chapter, the teacher opts for around four questions to provide a unified explanation, focusing on number system transformation calculations. In the second chapter, which involves primarily memory-based knowledge and features high question accuracy (over 50%), the teacher chooses one question with potential exam traps: the key distinction between ROM and RAM. Chapter 4 sees the teacher selecting two questions for feedback due to their complexity in calculations and reasoning these pertaining to binary tree computation and bubble sorting. Following this, the teacher encourages open student questioning, tailoring feedback accordingly. However, in larger classes, students tend to be reserved in posing questions, averaging only one to two questions per chapter.

Following the review phase, students in both groups took a learning performance test and completed intrinsic motivation and cognitive load scales. All the above data were used as posttest data for each chapter. In addition, the chatbot group was asked to evaluate the chatbot-based formative feedback, including its advantages and disadvantages. After receiving their

TABLE I
ATTRIBUTES OF THE EXPERIMENT'S CONTENT AND ASSESSMENT

Chapter	Learning units	Skills level required. (1: low 5: high)		Number of review questions (Test questions)	Post-test method
Chapter 1	1. Number system recognition 2. Number system conversion 3. Number system comparison 4. ASCII code recognition 5. ASCII code comparison 6. Character storage size	· Math skill · Memory · Comprehension · Application (comparison) · Analysis	5 3 3 3 1	19(13)	School-based computer test system
Chapter 2	1. Hardware system components 2. Differences in memory 3. CPU calculation 4. Hard disk capacity	· Math skill · Memory · Comprehension · Application · Analysis	1 5 2 1 1	9(9)	School-based computer test system
Chapter 4	1.Characteristics of the algorithm 2.Features of linear tables 3.Features of the stack 4.Binary tree calculation 5.Search and sort algorithm	· Math skill · Memory · Comprehension · Application (Judgment) · Analysis	5 4 5 5 3	11(9)	Website test system

[1]Teacher group: a. Students attend a face-to-face lecture. b. One week later, students do exercises on the testing platform. c. Students receive review scores. d. The teacher explains and gives feedback on some error-prone questions.
[2]Chatbot group: a. Students attend a face-to-face lecture. b. One week later, students do exercises on the chatbot via conversational interaction. c. Chatbot explains every question immediately if students want it to d. Students try to answer the question again if they want to e. Students receive scores.

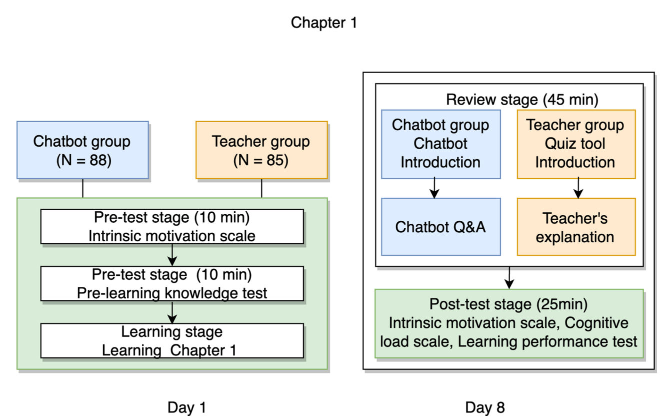

Fig. 4. Experiment procedure for Chapter 1.

feedback, the feedback design was updated and improved before experimenting with the second and fourth chapters. Due to space limitations, Fig. 4 illustrates the experiment process of Chapter 1 only. Chapters 2 and 4 follow the same procedure, except that the pretest stage is different.

D. Research Variables

1) Intrinsic Motivation: Participants completed the intrinsic motivation inventory [41], which contains 25 items, such as "When I review in this way, I always enjoy the process." Items were measured on a seven-point Likert scale from 1 (strongly disagree) to 7 (strongly agree). To assess the subjective experience of intrinsic motivation in an environment of formative feedback, we used the following subscales in this questionnaire: interest–enjoyment, tension–stress, perceived choice, perceived competence, and perceived value. There were seven items on the interest–enjoyment scale designed to assess intrinsic motivation. Perceived choice and perceived competence are positive predictors of intrinsic motivation, which contain four and five items, respectively. The tension–stress scale consists of five items that are negatively related to intrinsic motivation. Perceived value reflects people's internalization and self-regulation when experiencing useful activities. This scale contains four items. These five subscales were separately analyzed in this study, each being analyzed as a response variable.

2) Cognitive Load: The cognitive load questionnaire was adapted from McAuley et al. [41] and Sweller et al. [42]. This questionnaire contains eight items rated on a seven-point Likert scale that measures the dimensions of "mental load" and "mental effort." Mental load results from the interaction between task and subject characteristics (such as task structure and sequence of information). According to the Paas and Merrienboer model [43], the mental load can be determined based on our current understanding of the task and subject characteristics. Mental effort refers to how cognitive capacity is assigned to a particular task. The effort represents the cognitive input required to fulfill the requirements of the task and the degree of psychological input required. It should be considered when deciding how to organize and present learning material or how to guide students toward learning [44]. The "mental load" dimension consists of five questions, and the "mental effort" dimension consists of three questions.

3) Formative Feedback Exercises and Chapter Tests: A Professor from the School of Computer Science was asked to develop test questions based on the learning objectives in Chapters 1, 2, and 4. Question types include multiple-choice questions and fill-in-the-blank questions. For each correctly answered question, one point was awarded. To determine the initial level of the student's knowledge, we used the pretest at the start of each chapter. Following the completion of the review phase, the posttest was administered to gauge the student's level of mastery of the subject matter.

E. Data Analysis

To address the research questions, we first used a normality test to conduct reliability tests of all scales. Then, we conducted the linear mixed model (LMM) analysis for mean differences

TABLE II
RELIABILITY TESTS OF EACH DIMENSION OF INTRINSIC MOTIVATION AND COGNITIVE LOAD

Dimension	Mean	SD	Cronbach's alpha
Interest-enjoyment	4.917	1.03	0.848
Tension-stress	3.103	1.278	0.843
Perceived choice	4.74	1.033	0.64
Perceived competence	4.185	1.079	0.851
Perceived value	5.345	1.039	0.87
Mental load	3.59	1.157	0.89
Mental effort	3.821	1.09	0.805

TABLE III
LMM ABOUT INTRINSIC MOTIVATION

Parameter	Estimate	Std. Error	df	t	95% Confidence interval	
					Lower Bound	Upper Bound
Interest-enjoyment						
intercept	5.25	0.144	538.87	**36.55*****	4.97	5.53
time	-0.16	0.05	622.58	**-3.04****	-0.26	-0.06
Chatbot group	-0.55	0.2	538.87	**-2.71****	-0.94	-0.15
Chatbot group*time	0.19	0.07	622.58	**2.66****	0.05	0.33
Tension-stress						
intercept	2.91	0.16	536.65	**18.45*****	2.6	3.22
time	0.15	0.06	630.94	**2.74****	0.04	0.26
Chatbot group	0.48	0.22	536.65	**2.17***	0.05	0.91
Chatbot group*time	-0.26	0.08	630.94	**-3.25****	-0.41	-0.1
Perceived choice						
intercept	5	0.14	535.13	**36.748*****	4.74	5.27
time	-0.14	0.05	615.84	**-2.82****	-0.24	-0.04
Chatbot group	-0.41	0.19	535.13	**-2.13***	-0.78	-0.03
Chatbot group * time	0.14	0.07	615.84	**2.02***	0.004	0.27
Perceived competence						
intercept	4.13	0.16	536.63	**26.69*****	3.82	4.43
time	0.01	0.06	624.66	0.2	-0.1	0.12
Chatbot group	0.04	0.22	536.63	0.2	-0.38	0.47
Chatbot group * time	0.02	0.08	624.66	0.31	-0.13	0.18
Perceived value						
intercept	5.83	0.15	538.9	**38.16*****	5.53	6.13
time	-0.32	0.06	623.48	**-5.74*****	-0.42	-0.21
Chatbot group	-0.74	0.21	538.9	**-3.47****	-1.17	-0.32
Chatbot group * time	0.25	0.08	623.48	**3.26****	0.1	0.4

The teacher group is set to reference. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

in intrinsic motivation between groups. The method allowed for greater flexibility in the correlations between the repeated measures. In this analysis, intrinsic motivation was measured at four different times, it is treated as the repeated effect. For each dependent variable (interest–enjoyment, tension–stress, choice, competence, and value), groups representing different feedback types, time, and group × time interaction were treated as fixed effects, and subjects were treated as random effects with a variance components structure. Maximum likelihood estimation was used. Next, we conducted the independent sample t-test for the average cognitive load scores of the two groups after reviewing Chapters 1, 2, and 4. Last, we used a paired sample t-test to check whether the two groups' learning performance improved after the intervention. We further use the two-way analysis of covariance (ANCOVA) to determine whether the mean change in performance test scores between the two groups differed for each chapter. The fixed factors are "group" and "chapter," and the dependent variable is the posttest performance score, while the pretest performance score acts as the covariate. The result meets the homogeneity of regression slopes.

V. RESULTS

A. Reliability Tests

A normality test showed that all data conformed to a normal distribution. The Cronbach coefficient was calculated to test the internal consistency of the scale. The five dimensions of the intrinsic motivation scale and the two dimensions of the cognitive load scale and their Cronbach coefficient, means, and standard deviations are depicted in Table II.

B. Changes in Intrinsic Motivation Over Time

First, to verify the impact of chatbot-based formative feedback on intrinsic motivation over time, we first compare the initial intrinsic motivation scores of the two groups using an independent sample t-test. The results showed that the initial interest, perceived value, and choice scores of the chatbot group were significantly lower than those of the teacher group. There was no significant difference between initial perceived competence and stress. Then, we use the LMM for further analysis. Table III summarizes the result of the LMM for all dimensions of intrinsic motivation. As for interest, we found a significant main effect for the group ($t = -2.7$, $p < 0.01$) and time ($t = -3.04$, $p < 0.01$), and a significant interaction effect between the chatbot group and time ($t = 2.66$, $p < 0.01$). The teacher group had interest scores that were on average 0.55 more than the chatbot group, with both groups decreasing with time by about 0.16 points every time. However, the interest scores in the chatbot group tended to increase over time compared with the teacher group.

Second, when we compared tension–stress between the two groups, we found a significant main effect for the group ($t = 2.17$, $p < 0.05$) and time ($t = 2.74$, $p < 0.01$), and a significant interaction effect between the chatbot group and time ($t = -3.25$, $p < 0.01$). The chatbot group had stress scores that were on average 0.48 more than the teacher group, with both groups increasing with time by about 0.15 points every time. However, the stress scores in the chatbot group tended to decrease over time compared with the teacher group.

Third, we compared perceived choice between the two groups. We found a significant main effect for the group ($t = -2.13$, $p < 0.05$) and time ($t = -2.82$, $p < 0.01$), and a significant interaction effect between the chatbot group and time ($t = 2.02$, $p < 0.05$). The chatbot group had perceived choice scores that were on average 0.41 less than the teacher group, with both groups decreasing with time by about 0.14 points every time. However, the perceived choice scores in the chatbot group tended to increase over time compared with the teacher group.

Next, we compared perceived competence between the two groups. We observed that there was no significant main effect for the group ($t = 0.20$, $p > 0.05$) and time ($t = 0.20$, $p > 0.05$), and no significant interaction effect between the chatbot group and time ($t = 0.31$, $p > 0.05$). No heightening effect was observed for perceived competence. In addition, we found no significant improvement in perceived competence in the teacher group at any time. However, the perceived competence scores of the chatbot group at time 3 (Chapter 2) were significantly higher than that at time 1 (Chapter 1).

TABLE IV
COMPARISON OF COGNITIVE LOAD AFTER CHAPTER 1 WAS REVIEWED

Chapter 1	Group	*N*	Mean	SD	*t*	*F*	*p*
Cognitive load	Teacher	85	3.672	1.073	-0.052	0.778	0.959
	Chatbot	88	3.68	1.036			
Mental load	Teacher	85	3.558	1.224	-0.356	1.237	0.722
	Chatbot	88	3.621	1.095			
Mental effort	Teacher	85	3.863	1.139	0.497	1.239	0.62
	Chatbot	88	3.78	1.044			

TABLE V
COMPARISON OF COGNITIVE LOAD AFTER CHAPTER 2 WAS REVIEWED

Chapter 2	Group	*N*	Mean	SD	*t*	*F*	*p*
Cognitive load	Teacher	85	3.552	1.095	2.008	0.166	0.046
	Chatbot	88	3.203	1.183			
Mental load	Teacher	85	3.445	1.219	2.154	0.011	0.033
	Chatbot	88	3.046	1.218			
Mental effort	Teacher	85	3.729	1.07	1.519	2.042	0.131
	Chatbot	88	3.458	1.266			

Last, we compared the perceived value between the two groups, as given in Table III. We found a significant main effect for the group ($t = -3.47$, $p < 0.01$) and time ($t = -5.74$, $p < 0.001$), and a significant interaction effect between the chatbot group and time ($t = 3.26$, $p < 0.01$). The chatbot group had perceived value scores that were on average 0.47 less than the teacher group, with both groups decreasing with time by about 0.32 points every time. However, perceived value scores in the chatbot group tended to increase over time compared with the teacher group.

C. Differences in Cognitive Load

To measure the impact of two formative feedback methods on students' cognitive load, we measured cognitive load scores after each chapter was reviewed. The independent sample t-test of the cognitive load scores of the two groups after Chapter 1 is given in Table IV. It showed no significant difference in the total cognitive load scores of the two groups ($F = 0.778$, $p > 0.05$). The total cognitive load scores of both groups were low. The mean total score of the teacher group was 3.672 and the chatbot group was 3.680. The results showed that for the review of Chapter 1, the two formative feedback methods did not cause a large cognitive load on students.

The independent sample t-test of the cognitive load scores after the review of Chapter 2 is given in Table V. The data showed that the total cognitive load scores of the two groups decreased, and there is a significant difference ($F = 0.166$, $p < 0.05$). The total score of the chatbot group is smaller. Furthermore, the difference in cognitive load between the two groups is attributed to the mental load scores ($F = 0.011$, $p < 0.05$). As the two groups used the same exercises, there was no significant difference in mental effort ($F = 2.042$, $p > 0.05$).

Table VI displays the results of an independent sample t-test of the cognitive load scores for the two groups following the review of Chapter 4. The result showed that the total cognitive load scores of the two groups have increased, and there is a significant difference ($F = 0.008$, $p < 0.05$). The total cognitive load score for the teacher group reached a high of 3.96, whereas the total cognitive load score of the chatbot group was lower ($M = 3.523$). As for the mental load score, the score of the teacher group is high ($M = 3.92$). Regarding the mental effort score, the score of the chatbot group is smaller ($M = 3.508$), while the score of the teacher group ($M = 4.016$) has increased significantly with the more challenging content of Chapter 4.

TABLE VI
COMPARISON OF COGNITIVE LOAD AFTER CHAPTER 4 WAS REVIEWED

Chapter 4	Group	*N*	Mean	SD	*t*	*F*	*p*
Cognitive load	Teacher	85	3.956	1.2	2.523	0.008	0.013
	Chatbot	88	3.523	1.05			
Mental load	Teacher	85	3.92	1.317	2.078	0.018	0.039
	Chatbot	88	3.532	1.129			
Mental effort	Teacher	85	4.016	1.153	3.012	0.145	0.003
	Chatbot	88	3.508	1.065			

TABLE VII
POSTTEST RESULTS OF TWO-WAY ANCOVA

Chapter	Mean Difference Chatbot - Teacher	Std. Error	Sig.	95% Confidence Interval for Difference	
				Lower bound	Upper bound
Chapter 1	**2.427***	0.309	0	1.82	3.035
Chapter 2	0.247	0.309	0.425	-0.361	0.855
Chapter 4	**.815***	0.309	0.009	0.207	1.423

Based on estimated marginal means *. The mean difference is significant at the .05 level. Adjustment for multiple comparisons: Sidak.

D. Differences in Learning Performance

We used a paired sample t-test to check whether the performance of both groups improved after reviewing each chapter. The results showed that both groups significantly improved their performance test scores in each chapter through different feedback methods. We, then, used the independent samples t-test to compare whether there was a difference in the pretest performance scores of the three chapters between the two groups.

The results showed that the pretest performance scores of Chapters 1 and 2 of the chatbot group were significantly higher than those of the teacher group. There was no significant difference in the pretest performance scores of Chapter 4 between the two groups. One application of ANCOVA is to control for baseline (pretest) differences [45]. When groups differ in baseline, ANCOVA may be used to control these differences. To put it another way, posttest means for the groups are in comparison conditional on a particular pretest value. Therefore, to exclude the influence of pretest scores on the analysis results, we use the two-way ANCOVA to determine whether the mean change in test scores between the two groups differed for each chapter. The independent variables were group and chapter. Post-test scores served as the dependent variable. The pre-test scores were used as the covariate. The tests of between-subjects effects showed that "group" and "chapter" had a significant effect, and the interaction effect between "group" and 'chapter" is significant ($p < 0.001$). A simple effect analysis showed that as for Chapters 1 and 4, the chatbot group's post-test performance scores were higher than those of the teacher group after eliminating the influence of pre-test scores ($p < 0.05$). For Chapter 2, after eliminating the influence of pre-test scores on post-test scores, there is no significant difference between the learning performance of the two groups ($p > 0.05$). The results of the two-way ANCOVA are given in Table VII.

VI. Discussion

A. Chatbot-Based Formative Feedback Increases Students' Interest

As the current study showed, chatbot-based formative feedback can increase students' learning interest over time, as it can sustain students' attention over an extended period [46], [47]. The chatbot is designed as a personal assistant, responding to questions and providing immediate feedback to stimulate students' learning interests [48]. There is a question module included in the chatbot, which allows students to review each knowledge point easily and repeatedly. In addition, the chatbot includes rich media elements, such as text, static images, and dynamic images, which can help attract and retain students' attention. students' interests can be stimulated by the flexibility of learning [49]. Accordingly, the triggering of learning interest will increase the joy of learning [50].

Previous studies have suggested a novelty effect associated with chatbot-based learning, leading to heightened interest following a single intervention. However, there has been a lack of longitudinal validation. One potential rationale for students' prolonged interest retention in the current study could stem from their increasing familiarity with the chatbot-based dialogue format. This format resembles casual social conversations, offering flexibility in revisiting feedback. These factors could explain why chatbot-based formative feedback proves conducive for extended use, as evidenced in our study spanning 36 days—particularly beneficial for students with initially low interest in learning.

B. Chatbot-Based Formative Feedback Alleviates Students' Stress

According to this study, chatbot-based formative feedback can alleviate student stress over time. It is plausible that students who receive chatbot-based feedback enjoy a more relaxed atmosphere, because the review is student centered and largely self-paced, and the interactive process encourages them to actively participate [51]. In teacher-based feedback, the teacher exerts control over the reviewing progress of students. For example, he or she controls the number and time of questions according to his or her teaching experience, which makes it difficult to meet the learning needs of each student. In addition, students must be able to answer questions at any time during the review period [52], which results in a high cognitive load on students, leading to tension or anxiety, especially in a large class. Under such conditions, students' motivation and performance will decline [53]. Consequently, this study suggests chatbots can be used as a feedback tool before exams to relieve tension and stress caused by exams. Continuous use of chatbot-based feedback can help relieve stress.

C. Students May Feel More in Control of Their Learning Journey With the Chatbot

Chatbot-based formative feedback can potentially enhance students' perceived autonomy over time. This is attributed to the self-paced nature of the chatbot-based approach, which provides students with increased control over their learning experience as opposed to being subjected to standard teaching methods by teachers. First, students can try to do the same question again if they answer incorrectly. Second, students can ask the chatbot to give a detailed analysis if they want. Third, students can control how long they review each question. Consequently, the use of chatbot-based formative feedback can cultivate greater student autonomy, bolstering their ability to take charge of their learning. Despite initial low levels of perceived choice, the chatbot group's sense of autonomy increased over time through continuous interaction with the chatbot. This finding highlights the importance of providing autonomous learning opportunities through chatbots for students who initially perceive limited choice [54]. Such continuous intervention can promote a greater sense of choice, leading to enhanced intrinsic motivation [55]. This underscores the significance of instructors prioritizing the integration of chatbots to support student autonomy and foster intrinsic motivation.

D. Perceived Competence Over Time is Similar for Both Groups

The findings indicated no significant difference in perceived competence between the two groups over time. This suggests that both groups possessed similar levels of self-efficacy in terms of their ability to manage and comprehend the learning content to achieve desired outcomes. Notably, the chatbot-based design did not result in any noticeable increase in perceived competence over the traditional approach. Perceived competence is a subjective evaluation of an individual's own competence, which is influenced by their beliefs, attitudes, and experiences related to the task [32]. Given that both groups were provided with identical learning material that required the same abilities and skills to comprehend, there was no significant difference in their self-assessed perceived competence. As a result, the chatbot-based intervention did not have a discernible impact on students' perceived competence over the traditional approach.

E. Chatbot-Based Formative Feedback Increases Students' Perceived Value

The chatbot-based formative feedback significantly heightens perceived value over time compared with teacher-based formative feedback. We extend the past research that new technological innovation needs to examine the novelty effect as the perceived value may decrease over time [47]. We establish the sustainable effects of chatbot-based formative feedback on students' perceived value. A possible explanation is that students find the flexible options of this learning technology useful over time. Following the technology acceptance model, students who perceive the chatbot-based formative feedback useful are more likely to accept them [56]. Therefore, if the design is concise and easy to use, teachers can confidently integrate chatbots into the classroom to provide formative feedback.

F. Chatbot-Based Formative Feedback can Reduce Students' Cognitive Load

It was found that chatbot-based formative feedback could reduce students' cognitive load on conceptual and difficult knowledge. According to Table I, the skill required for Chapter 1 is high math skills, and the required cognitive load depends on students' computing level. For this chapter, both feedback methods do not show a significant difference. They both explain how to do the calculations but do not make the students better at math.

However, Chapter 2 primarily relies on memory and the student–chatbot interactions differ from traditional teacher–student dynamics. The chatbot's interactive feature helps students practice exercises repeatedly, aiding memory retention and concept acquisition while reducing cognitive load, especially mental strain. The chatbot formative feedback approach, which consists of progressive prompts, provides a scaffolding approach that facilitates students' mastery of the learning material and relieves their cognitive burden [57]. In addition, the approaches to acquiring knowledge and receiving feedback are different. Students in the teacher group employed the "Questionnaire Star" for answering questions, potentially leading to frustration due to the lack of immediate feedback after responses. This circumstance could result in heightened cognitive strain. On the contrary, the chatbot group received instant feedback, effectively reducing their learning stress and mental load.

Chapter 4 involves skills in mathematics, comprehension, and practical application. In terms of challenging concepts, feedback from the teacher that is hard to grasp and follow can lead to increased mental strain for students. Furthermore, differences in cognitive effort can be attributed to the impact of the chatbot. Through interactive dialogues, the chatbot can decrease the mental effort that students need to invest in organizing information and completing learning tasks [58]. This advantage might arise from the chatbot's prompt response to student answers and the provision of additional learning opportunities, while formative feedback from the teacher involves a delay and less flexibility and control.

G. *Chatbot-Based Formative Feedback Promotes Students' Mastery of Applied Knowledge*

This study demonstrated that chatbots can provide students with better formative feedback to promote their mastery of applied knowledge. As shown in Section V, students in the chatbot group scored higher than those in the teacher group on Chapters 1 and 4 post-tests. This result is attributed to the design of the conversational interactive feedback, which allows students to review based on their progress, receive immediate feedback, and repeat the exercises if needed. This facilitates the formation and retention of knowledge. With repeated practice, students will learn how to regulate their cognitive and metacognitive strategies and progress to master the learning concept and problem-solving technique. These options; however, are limited in the traditional review setting. For a more conceptual learning unit where students must memorize, a chatbot can potentially facilitate their knowledge mastery just as effectively as teachers in this study's context. As the memory schema required by this type of knowledge is relatively simple, related concepts are structurally linked together to reduce the load on working memory [47]. Therefore, this study suggests that chatbots could be more effective if used as a feedback tool for applied knowledge accompanied by a more active method of learning-centric feedback.

H. *Implications of This Study*

First, despite chatbots being used as formative feedback tools, little is known about their sustainable influence on learning performance and psychological factors. Our study responds to this gap by offering empirical evidence on the sustainable impact of chatbots on intrinsic motivation, cognitive load, and learning performance. Specifically, the study was designed to overcome a one-off measure, thereby mitigating the influence of novelty. Throughout the 36-d duration, the positive effects of the chatbot were sustained.

Second, we propose six design principles based on SDT, CLT, and feedback framework. Our results validate the efficacy of certain design principles. Specifically, DP1 was observed to positively impact students' interests and perceived choices. Thus, for future developments of chatbot-based formative feedback, the adoption of DP1 could be beneficial. Furthermore, the incorporation of additional multimedia components (such as voice and video) and personalized features (such as adaptability) could be considered.

Our findings highlight the enduring impact of DP3 and DP4 in lessening cognitive load. When students provide incorrect answers, they receive hints, guiding them to focus solely on essential content. This process aids in easing students' cognitive burden. While our outcomes indicate no immediate reduction in cognitive load following the review of Chapter 1, subsequent time points reveal positive effects. This suggests that continued application of this principle can yield benefits. It is worth noting that chatbots should be designed with a streamlined conversation flow. Notably, in the experiment's description, a distinction between the two groups is the chatbot's concise and direct communication compared with the teacher's approach. We also find that DP5 and DP6 positively affect learning performance, especially for applied and difficult knowledge. The feedback for each question includes an evaluation section and an information section. It does not give a detailed analysis directly. This approach promotes independent thinking among students when grappling with complex issues and discourages overreliance on explanations. In the future, we can focus on applying these two design principles to more difficult knowledge. Finally, we found that DP2 failed to increase students' perceived competence as the students in the teacher group also received the same questions. This design principle does not seem to yield greater advantages using chatbots.

I. *Limitations and Future Research*

In this longitudinal study, we deployed a chatbot as a formative feedback tool, demonstrating the potential of such a tool. Future research can improve this preliminary study by addressing its limitations.

First, although the current chatbot-based feedback system provides students with adequate ability and autonomy support, this study disregards the collaborative interaction between students as well as the necessity for students to communicate closely with others. Because our chatbot is rule based, under this circumstance, it would be beneficial if it could act as a bridge for student collaboration. Enabling group members to assist one another and even to "teach" the chatbot how to solve problems.

If this limitation can be overcome, this tool may improve the relevance dimension of students' intrinsic motivation and increase the effectiveness of chatbots in the classroom.

Second, our chatbot's capability for assisting students with problem-solving needs improvement. Currently, students are restricted by the answers provided by the question-and-answer knowledge base. When they face a problem or are unclear about the feedback, they are unable to seek further clarification and are more likely to fall into a quagmire of uncertainty. A potential solution could be to use more advanced natural language processing technologies, such as large language models, to enhance the chatbot's ability to understand and respond to complex questions.

Finally, our chatbot needs to be more personalized. The formative feedback model of Narciss and Huth [2] and Hattie and Timperley [33] suggests that the form, content, and function of feedback should be personalized to the elements of the teaching situation, as well as the learner's characteristics. The learner's characteristics include learning goals, previous knowledge level, level of motivation to learn, cognitive load level, and the method of choosing feedback, such as whether a more detailed analysis is necessary and whether it is necessary to try to solve the problem again. The teaching context includes objectives, content, and learning activities. These aspects mentioned above, however, have not been fully implemented. The chatbot provides feedback based on the correctness of the answer and does not adapt to the individual's learning characteristics. Future research may explore adaptive strategies that overcome these limitations.

VII. Conclusion

In this study, we used a longitudinal approach to demonstrate the potential of a chatbot to provide sustained formative feedback, improve intrinsic motivation and learning performance, as well as alleviate the cognitive load. The use of chatbots is increasing across a wide range of fields, including education. Improving the usefulness of these resources as learning tools could have a profound effect on the learning process and the effectiveness of teaching. An encouraging finding from the study is that the chatbot-based formative feedback design can stimulate important dimensions of intrinsic motivation. First, a chatbot-based learning can attract students' attention and increase their perceived choice over an extended duration. Moreover, interacting with a chatbot has the potential to sustainably reduce cognitive load and learning pressure as well as equally effective as teachers in enhancing students' perceived competence. In addition, formative feedback delivered through chatbots can help alleviate students' cognitive load when grappling with conceptual and challenging knowledge. Moreover, the use of chatbots can lead to a more profound mastery of applied knowledge.

Although teacher–student interactions are not easily replicated by the Flow.ai chatbot design tool, our study confirmed that this tool can enhance teachers' feedback, enriching students' learning performance and experiences. This is particularly relevant in the age of advanced AI-like ChatGPT, which offers superior performance but might pose challenges in maintaining control over its responses. The Flow.ai tool, on the other hand, can be tailored and manipulated by the instructional designers. Future research could explore techniques for effectively utilizing ChatGPT to provide valuable formative feedback.

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Jiaqi Yin received the master's degree in educational technology from Hubei University, Wuhan, China, in 2021. She is currently working toward the Ph.D. degree in intelligent education from East China Normal University, Shanghai, China.

Her major research interests include chatbot-based learning, metacognitive instruction, and learning analysis.

Tiong-Thye Goh received the Ph.D. degree in information systems from Massey University, New Zealand, in 2007.

He is currently a Senior Lecturer in information management with the Victoria University of Wellington, Wellington, New Zealand. His publications appeared in *Journal of Business Research*, *Journal of Information Systems Education*, *Commerce Research*, *British Journal of Educational Technology*, and *Journal of Educational Technology & Society*. His research interests include social media, emotion computing, classification, education technology, analytics, and user behavior.

Yi Hu received the Ph.D. degree in developmental and educational psychology from East China Normal University, Shanghai, China, in 2004.

He is currently a Full Professor with the School of Psychology and Cognitive Science with East China Normal University. His research focuses on the design of human–machine learning environment design.